

SENTENCIAS DML EN SQL

¿Qué es DML?

DML (Data Manipulation Language) es un conjunto de instrucciones del lenguaje SQL que se utilizan para manipular la información almacenada en una base de datos.

Estas sentencias permiten trabajar directamente con los datos, es decir, agregar, consultar y eliminar registros dentro de las tablas.

A diferencia de otras partes de SQL (como DDL, que crea tablas), DML se enfoca en modificar el contenido de las tablas ya existentes.

Principales sentencias DML

INSERT

La sentencia INSERT se utiliza para agregar nuevos registros (filas) a una tabla.

- Permite ingresar datos específicos en cada columna
 - Es fundamental para almacenar nueva información
 - Se debe respetar el tipo de dato de cada campo
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SELECT

La sentencia **SELECT** se utiliza para **consultar o visualizar los datos** almacenados en una tabla.

- Permite ver todos los registros o solo algunos
 - Se pueden seleccionar columnas específicas
 - Se puede usar la cláusula WHERE para filtrar resultados
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DELETE

La sentencia **DELETE** se utiliza para **eliminar registros** de una tabla.

- Se debe usar con la condición WHERE para evitar eliminar todos los datos
 - Permite borrar uno o varios registros según un criterio
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APLICACIÓN AL MODELO DE DATOS

1. Tabla: security.role

INSERT

-- Inserta un nuevo rol en la tabla

```
INSERT INTO security.role(name, description, created_at, state)
VALUES ('Admin','Administrador del sistema', NOW(),'ACTIVE');
```

-- Inserta otro rol

```
INSERT INTO security.role(name, description, created_at, state)
VALUES ('User','Usuario general', NOW(),'ACTIVE');
```

-- Inserta un rol con estado inactivo

```
INSERT INTO security.role(name, description, created_at, state)
VALUES ('Guest','Invitado', NOW(),'INACTIVE');
```

SELECT

-- Muestra todos los registros de la tabla

```
SELECT * FROM security.role;
```

-- Muestra solo el nombre y la descripción

```
SELECT name, description FROM security.role;
```

-- Muestra solo los roles activos

```
SELECT * FROM security.role WHERE state = 'ACTIVE';
```

DELETE

-- Elimina el rol Guest

```
DELETE FROM security.role WHERE name = 'Guest';
```

-- Elimina el rol User

```
DELETE FROM security.role WHERE name = 'User';
```

-- Elimina el rol Admin

```
DELETE FROM security.role WHERE name = 'Admin';
```

2. Tabla: security.user

INSERT

-- Inserta un usuario

```
INSERT INTO security."user"(username,email,password,role_id,created_at,state)
VALUES ('eric','eric@mail.com','1234',(SELECT id FROM security.role LIMIT
1),NOW(),'ACTIVE');
```

-- Inserta otro usuario

```
INSERT INTO security."user"(username,email,password,role_id,created_at,state)
VALUES ('juan','juan@mail.com','1234',(SELECT id FROM security.role LIMIT
1),NOW(),'ACTIVE');
```

-- Inserta otro usuario

```
INSERT INTO security."user"(username,email,password,role_id,created_at,state)
VALUES ('ana','ana@mail.com','1234',(SELECT id FROM security.role LIMIT
1),NOW(),'ACTIVE');
```

SELECT

-- Muestra todos los usuarios

```
SELECT * FROM security."user";
```

-- Muestra solo usuario y correo

```
SELECT username, email FROM security."user";
```

-- Muestra usuarios activos

```
SELECT * FROM security."user" WHERE state = 'ACTIVE';
```

DELETE

-- Elimina el usuario ana

```
DELETE FROM security."user" WHERE username = 'ana';
```

-- Elimina el usuario juan

```
DELETE FROM security."user" WHERE username = 'juan';
```

-- Elimina el usuario eric

```
DELETE FROM security."user" WHERE username = 'eric';
```

3. Tabla: inventory.category

INSERT

-- Inserta categorías

```
INSERT INTO inventory.category(name, description, created_at, state)
VALUES ('Electronics','Dispositivos electrónicos', NOW(),'ACTIVE');

INSERT INTO inventory.category(name, description, created_at, state)
VALUES ('Food','Alimentos', NOW(),'ACTIVE');

INSERT INTO inventory.category(name, description, created_at, state)
VALUES ('Clothes','Ropa', NOW(),'ACTIVE');
```

SELECT

-- Muestra todas las categorías

```
SELECT * FROM inventory.category;
```

-- Muestra solo nombres

```
SELECT name FROM inventory.category;
```

-- Muestra categorías activas

```
SELECT * FROM inventory.category WHERE state = 'ACTIVE';
```

DELETE

-- Elimina categorías

```
DELETE FROM inventory.category WHERE name = 'Clothes';

DELETE FROM inventory.category WHERE name = 'Food';

DELETE FROM inventory.category WHERE name = 'Electronics';
```

4. Tabla: inventory.product

INSERT

-- Inserta productos

```
INSERT INTO inventory.product(name, description, price, category_id, created_at,
state)
VALUES ('Laptop','Portátil',2000,(SELECT id FROM inventory.category LIMIT
1),NOW(),'ACTIVE');
```

```
INSERT INTO inventory.product(name, description, price, category_id, created_at, state)
```

```
VALUES ('Phone','Celular',1000,(SELECT id FROM inventory.category LIMIT 1),NOW(),'ACTIVE');
```

```
INSERT INTO inventory.product(name, description, price, category_id, created_at, state)
```

```
VALUES ('Tablet','Tableta',800,(SELECT id FROM inventory.category LIMIT 1),NOW(),'ACTIVE');
```

SELECT

```
-- Muestra todos los productos
```

```
SELECT * FROM inventory.product;
```

```
-- Muestra nombre y precio
```

```
SELECT name, price FROM inventory.product;
```

```
-- Muestra productos con precio mayor a 900
```

```
SELECT * FROM inventory.product WHERE price > 900;
```

DELETE

```
-- Elimina productos
```

```
DELETE FROM inventory.product WHERE name = 'Tablet';
```

```
DELETE FROM inventory.product WHERE name = 'Phone';
```

```
DELETE FROM inventory.product WHERE name = 'Laptop';
```

5. Tabla: inventory.inventory

INSERT

```
-- Inserta registros de inventario
```

```
INSERT INTO inventory.inventory(product_id, quantity, created_at, state)
```

```
VALUES ((SELECT id FROM inventory.product LIMIT 1),10,NOW(),'ACTIVE');
```

```
INSERT INTO inventory.inventory(product_id, quantity, created_at, state)
```

```
VALUES ((SELECT id FROM inventory.product LIMIT 1),20,NOW(),'ACTIVE');
```

```
INSERT INTO inventory.inventory(product_id, quantity, created_at, state)
VALUES ((SELECT id FROM inventory.product LIMIT 1),30,NOW(),'ACTIVE');
```

SELECT

-- Muestra inventario

```
SELECT * FROM inventory.inventory;
```

-- Muestra cantidades

```
SELECT quantity FROM inventory.inventory;
```

-- Muestra cantidades mayores a 15

```
SELECT * FROM inventory.inventory WHERE quantity > 15;
```

DELETE

-- Elimina registros por cantidad

```
DELETE FROM inventory.inventory WHERE quantity = 30;
```

```
DELETE FROM inventory.inventory WHERE quantity = 20;
```

```
DELETE FROM inventory.inventory WHERE quantity = 10;
```

6. Tabla: bill.bill

INSERT

-- Inserta facturas

```
INSERT INTO bill.bill(user_id, total, created_at, state)
```

```
VALUES ((SELECT id FROM security."user" LIMIT 1),5000,NOW(),'ACTIVE');
```

```
INSERT INTO bill.bill(user_id, total, created_at, state)
```

```
VALUES ((SELECT id FROM security."user" LIMIT 1),3000,NOW(),'ACTIVE');
```

```
INSERT INTO bill.bill(user_id, total, created_at, state)
```

```
VALUES ((SELECT id FROM security."user" LIMIT 1),1000,NOW(),'ACTIVE');
```

SELECT

-- Muestra facturas

```
SELECT * FROM bill.bill;
```

-- Muestra totales

```
SELECT total FROM bill.bill;
```

-- Muestra facturas mayores a 2000

```
SELECT * FROM bill.bill WHERE total > 2000;
```

DELETE

-- Elimina facturas

```
DELETE FROM bill.bill WHERE total = 1000;
```

```
DELETE FROM bill.bill WHERE total = 3000;
```

```
DELETE FROM bill.bill WHERE total = 5000;
```

7. Tabla: bill.bill_item

INSERT

-- Inserta detalles de factura

```
INSERT INTO bill.bill_item(bill_id, product_id, quantity, unit_price, total,  
created_at, state)
```

```
VALUES ((SELECT id FROM bill.bill LIMIT 1),(SELECT id FROM inventory.product  
LIMIT 1),2,1000,2000,NOW(),'ACTIVE');
```

```
INSERT INTO bill.bill_item(bill_id, product_id, quantity, unit_price, total,  
created_at, state)
```

```
VALUES ((SELECT id FROM bill.bill LIMIT 1),(SELECT id FROM inventory.product  
LIMIT 1),1,800,800,NOW(),'ACTIVE');
```

```
INSERT INTO bill.bill_item(bill_id, product_id, quantity, unit_price, total,  
created_at, state)
```

```
VALUES ((SELECT id FROM bill.bill LIMIT 1),(SELECT id FROM inventory.product  
LIMIT 1),3,500,1500,NOW(),'ACTIVE');
```

SELECT

-- Muestra detalles

```
SELECT * FROM bill.bill_item;
```

-- Muestra cantidad y total

```
SELECT quantity, total FROM bill.bill_item;
```

-- Muestra registros con total mayor a 1000

```
SELECT * FROM bill.bill_item WHERE total > 1000;
```

DELETE

-- Elimina registros

```
DELETE FROM bill.bill_item WHERE total = 800;
```

```
DELETE FROM bill.bill_item WHERE total = 1500;
```

```
DELETE FROM bill.bill_item WHERE total = 2000;
```